NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

DEPARTMENT OF ELECTRICAL ENGINEERING

Mid-Semester Exam-1 Roll No.....

Subject: Network Synthesis & Filters Subject code: EEPC205

3rd Sem (Electrical Engg) Max marks: 20

Time allotted: 50 mins

All questions are compulsory

Q1. Check whether the following network functions are positive real functions or not & apply the Sturm's test wherever required.

(a) $F(s) = \dfrac{3s^2 + 5}{s(s^2 + 1)}$ 2

(b) $F(s) = \dfrac{s^2 + \frac{3}{4}s + \frac{3}{4}}{s^2 + s + 4}$ 2

Q2. Check whether the following function is Hurwitz or not

$$H(s) = s^5 + s^3 + s$$ 4

Q3. Which type of elements the following impedance function include & then synthesize it in Foster-I form.

$$Z(s) = \frac{s(s+2)(s+5)}{(s+1)(s+4)}$$ 4

Q4 Find the Causer-I form for the following driving point LC impedance function

$$Z(s) = \frac{6s^3 + 2s}{9s^4 + 4s^2 + \left(\frac{1}{6}\right)}$$ 4

Q5. Comment which type of elements the following admittance function can include and then realize it (using any method)

$$Y(s) = \frac{s^2 + 5s + 6}{s^2 + 5s + 4}$$ 4